AWANTIKA SRIVASTAVA

Data Scientist | Data Analytics | Machine Learning

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PROFILE SUMMARY

Data Scientist with **2+ years of experience in data analytics, statistical modeling, and machine learning**. Skilled in **SQL, Python, statistical analysis, and data visualization** to generate data-driven insights and support business decision-making. Experienced in **data extraction, exploratory data analysis, predictive modeling, and data quality validation** across large datasets. Strong ability to **identify patterns, interpret analytical results, and communicate insights to stakeholders**.

CORE TECHNICAL SKILLS

- **Programming & Data Handling:** Python, Numpy, Pandas, Matplotlib, Scikit-learn, C++
- **Statistics & Mathematics:** EDA, Statistical Modeling, Hypothesis Testing, A/B Testing, Probability, Linear Optimization, Statistical Analysis, Data Cleaning
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, Hyperparameter Tuning, Feature Engineering
- **Deep Learning:** Neural Networks, CNN, RNN, LSTM, BERT, Vision Transformers, Self-Attention, Cross-Attention, Sequence Modeling, TensorFlow, PyTorch, Keras, Time-Series Forecasting, ARIMA, SARIMA
- **Computer Vision**: Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Feature Extraction, Multi-Modal Fusion Concepts
- **NLP**: Text preprocessing, Tokenization, Chunkings, Transformers, Sentiment Analysis, Semantic Search, Transformer-based models, NLTK, SciPy
- **Data Visualization:** PowerBI, Matplotlib, Seaborn
- **Databases & Tools:** MySQL, PostgreSQL, Git, Linux
- **Concepts:** Data Mining, Predictive Modeling, Pattern Analysis, Business Analytics

EXPERIENCE

Data Scientist | PPS International Pvt. Ltd. **January 2024-Present**

- Performed **data extraction and querying using SQL** to gather structured datasets for analytics and modeling tasks.
- Built **data analysis pipelines using Python (Pandas, NumPy)** to clean, preprocess, and transform large datasets.
- Applied **statistical analysis and machine learning techniques** such as **Regression, Decision Trees, and Random Forest** to identify patterns and generate predictive insights.
- Conducted **data validation, data quality checks, and error detection** to ensure reliability and accuracy of analytical outputs.
- Developed **analytical models and reports** to support data-driven business decision making.
- Created **data visualizations and dashboards** to present insights and trends to internal stakeholders.
- Performed **pattern identification and trend analysis** to uncover actionable insights from complex datasets.
- Collaborated with **cross-functional teams including analytics engineers and business stakeholders** to define analytical methodologies and deliver insights.
- Conducted **data troubleshooting and anomaly detection** to identify inconsistencies and improve data integrity.

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- Designed and **deployed a real-time computer vision**–based ML system to detect unsafe driver behaviors from continuous video streams.
- Trained and **optimized CNN-based** object detection **models (SSD MobileNet** architecture) to perform real-time inference on video data.
- **Implemented end-to-end ML pipelines** covering data ingestion, preprocessing, model training, evaluation, and production inference.
- Achieved **20–25 FPS** real-time processing with **<150 ms** inference latency by optimizing models for **deployment**.
- Built and integrated **a Flask-based** web **interface** to **visualize** detections and automatically recorded 30-second event clips, reducing manual **review** effort.

LLM & RAG-Based AI Chatbot | Applied ML Project

- Built a Lightweight **LLM-Powered chatbot** using **TinyLLaMA** to answer user queries over content.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve relevant resume sections for contextual question answering.
- Selected **TinyLLaMA** to ensure **low memory footprint and fast inference**, making the solution suitable for resource-constrained environments.
- Deployed the chatbot as an interactive **Streamlit web application** for real-time user interaction.

YouTube Comments Sentiment Analyzer | Link -https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/

- Deployed transformer-based **NLP models** as **production-ready** services with **REST APIs**.
- **Fine-tuned** and served a **DistilBERT**-based **sentiment** classification model for large-scale text inference.
- **Built** and deployed an interactive **streamlit web application** to perform real-time **sentiment analysis** on YouTube comments.
- **Processed** high-volume text **data** with sub-second inference **latency** for real-time sentiment analysis.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**

Bachelor of Technology